

1. Data Preprocessing

- File Conversion: Converted training data from XLSX to CSV format and added headers.
- Data Exploration: Analyzed emotion class frequencies; noted imbalance with 'HAPPY' most frequent and 'FEAR' least frequent.
- Normalization: Used `hazm.Normalizer` to standardize Persian text by fixing spaces, removing diacritical marks, and converting numbers.
- Cleaning: Removed URLs, HTML codes, punctuations, emojis, and digits.
- Tokenization: Split text into tokens using `hazm.Tokenizer`.
- Stop Words Removal: Eliminated common words to reduce dimensionality and focus on meaningful content.
- Vectorization: Applied `TfidfVectorizer` with n-gram range (1, 3) and specified parameters to transform text into TF-IDF matrix.
- Label Encoding: Converted categorical emotion classes into numerical values.

2. Tree-Based Methods

- Evaluation Metrics: Used precision, recall, F1 score, and accuracy for model evaluation.
1. Decision Tree:
 - Initialization: Trained a decision tree classifier.
 - Hyperparameter Tuning: Used `GridSearchCV` for best parameters.
 - Results: Accuracy of 39.49%, with precision, recall, and F1 score all around 0.39.
 2. Random Forest:
 - Initialization: Trained a random forest classifier.
 - Hyperparameter Tuning: Applied `GridSearchCV` for optimization.

- Results: Accuracy of 52.39%, with precision of 0.60, recall of 0.52, and F1 score of 0.47.
- 3. Gradient Boosting:
 - Initialization: Trained a gradient-boosting classifier.
 - Hyperparameter Tuning: Faced computational challenges with GridSearchCV.
 - Results: Accuracy of 55.84%, with precision, recall, and F1 score all around 0.56.
- 4. XGBoost:
 - Initialization: Trained an XGBoost classifier.
 - Hyperparameter Tuning: Encountered computational limits with RandomizedSearchCV.
 - Results: Accuracy of 56.85%, with precision, recall, and F1 score all around 0.56.

3. Final Model Evaluation

- Logistic Regression:
 - Performance: Achieved 62.13% accuracy using 5-fold cross-validation.
- Support Vector Classifier (SVC):
 - Performance: Achieved 62.64% accuracy using 5-fold cross-validation.
- Ensemble Learning:
 - Approach: Combined Logistic Regression and SVC with soft voting.
 - Outcome: Marginal accuracy improvement; used for final predictions.

4. Test Set Handling

- Prediction: Generated predictions for test data and saved results in a CSV file under the header 'Y', with 'X' containing raw sentences.